CODEFORCES
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HOME TOP CATALOG CONTESTS GYM PROBLEMSET GROUPS RATING EDU API CALENDAR HELP

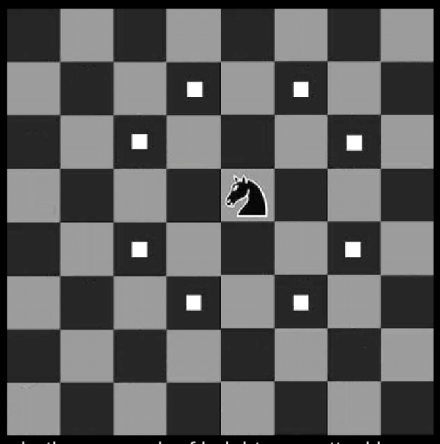
Codeforces Round 1133 (Div. 1, Div. 2) is coming soon! Register here.

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

G. The Knights of Königsberg

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a chessboard of size $N \times M$. There are K knights placed on distinct cells of this board. The standard movement of a knight in chess is as follows: it moves in an "L" shape —either two squares in one direction and then one square perpendicular to that, or one square in one direction and then two squares perpendicular to that. A knight can jump over other pieces.



Your task is to determine whether any pair of knights are attacking each other. If at least one pair of knights are in positions such that one can attack the other in a single move, print "YES". Otherwise, print "NO".

Input

The first line contains three integers N, M, and K ($1 \le N, M \le 1000, 0 \le K \le 10^5$) — the number of rows, the number of columns, and the number of knights on the board.

Each of the next K lines contains two integers x_i, y_i ($1 \le x_i \le N, 1 \le y_i \le M$) — the position of the i-th knight on the board.

It is guaranteed that all K knights are placed on distinct positions.

Examples

input Copy

```
6 4 4
2 3
4 2
1 2
3 3
```

output Copy

```
YES
```

input Copy

CSE221 Assignment 04 Fall 2026
Contest is running
8 days
Contestant

→ Submit?

Language: Java 21 64bit
Choose file: Choose File No file chosen
Success!
CLOUDFLARE
Privacy • Help
Submit

→ Last submissions

Submission	Time	Verdict
397690099	Nov/07/2026 09:53	Accepted

```
4 4 4
1 1
1 3
2 2
2 4
```

output Copy

```
NO
```

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The only programming contests Web 2.0 platform
Server time: Nov/07/2026 15:05:49 UTC+6 (f1).
mobile version
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